Predicting Card Mana Value in the MTG Star Trek Set using Random Forest and Text Analysis

PACE Strategy Document

Primary objective:

To find insights from the Magic The Gathering Star Trek set data that can showcase the skills I've acquired completing the Google Advanced Data Analytics Professional Certificate.

This project should build a predictive machine learning model to determine the underlying mana pricing formula for card mana values in *Magic: The Gathering*. Utilizing a raw JSON dataset pulled from the Scryfall API containing the complete *Star Trek* (TRK) set to extract feature data from card ability text alongside base stats like power and toughness. These features are fed into a "Random Forest" classification model to predict the balanced mana cost of a given card.

Technical Skills

- Programming Languages: Python
- Platforms & Tools: Tableau
- Python Packages: numpy, Pandas, Scipy, seaborn, Matplotlib, statsmodels, scikit-learn
- Advanced Statistics: Hypothesis testing, confidence intervals, and ANOVA (Analysis of Variance)
- Analytical Workflows: Exploratory Data Analysis (EDA) and data storytelling or stakeholder communication
- Machine Learning Concepts: Unsupervised learning (specifically K-means clustering) and Feature Engineering
- Machine Learning Models: regression (linear, logistic), Naive Bayes, decision trees, random forest, AdaBoost, XGBoost

Scope and boundaries:

Here is exactly how an MTGA project maps to and exceeds the Salifort capstone:

The Problem: Salifort asked to predict employee churn (a Yes/No answer) for HR stakeholders; this project predicts a card's mana cost (a multi-class answer: 1, 2, 3, etc.) for game balance stakeholders.

The Data: Salifort used a perfectly pre-cleaned, static CSV file with 15,000 rows from Kaggle; this project uses raw, real-world JSON data pulled directly from the Scryfall API, proving I can handle messy, unstructured data.

The Modeling: Salifort required you to build Logistic Regression and Random Forest models to find feature importance; my project uses the exact same Random Forest architecture to evaluate how heavily power, toughness, and abilities weigh on the final cost.

The Bonus (Standing Out): Salifort gave you all the variables ready to go (like hours worked or evaluation scores); my project requires me to create my own variable by having Python count the words or extract keywords from the card's ability text.

Business Understanding:

Stakeholders (conceptually): Lead Game Designers, the Play Design Team, and Product Managers (e.g., Wizards of the Coast).

Business Problem: Maintaining game balance is critical to player retention and product longevity. If cards are designed underpowered, a new set fails to make an impact; if cards are overpowered, they warp the competitive meta and require emergency bans, damaging consumer trust.

Project Value: This predictive model serves as an automated balancing tool for future set development. By determining the baseline mana value for complex abilities and stats, the design team can proactively

prevent power creep, ensure consistent card pricing, and better forecast which cards have the mathematical efficiency to see wider competitive meta play outside of Limited formats.

Data Understanding:

Data Source: The project utilizes the bulk JSON dataset pulled from the Scryfall API, filtered specifically for the Magic: The Gathering Star Trek (TRK) set releasing in November 2026. It initially looks like Scryfall will have the data required for this project, but I have yet to view the data and may need to find another source.

Data Limitations: The dataset contains purely static, printed attributes. It lacks actual gameplay data, win rates, or meta-share statistics. The model must assume the designers perfectly balanced the mana cost, which is not always historically accurate.

Unstructured Text: The card abilities are written in unstructured natural language rather than structured categorical variables. Extracting meaning requires Natural Language Processing (NLP), highly complex or unique mechanics may be oversimplified by the algorithm.

Synergy Blindness: The dataset evaluates each card as an isolated entity. The model cannot account for combo potential or set-wide mechanical synergies (such as the Federation keyword scaling with other cards) that heavily influence a card's true value.

Modeling and Evaluation:

Model: Random Forest algorithm to handle the complex, non-linear relationships between a card's unstructured text and its stats.

Evaluation Metrics:

- Mean Absolute Error (MAE) to measure the average distance of incorrect predictions (e.g., off by 0.5 mana).
- Accuracy, Precision, Recall, F1-Scores to track the exact match rate.

Conclusion:

Final Deliverable: A Python pipeline that extracts and processes Scryfall API JSON data, trains a predictive model, and exports the findings to an interactive Tableau dashboard for stakeholder presentation.

Future Steps: If the model successfully predicts standard mana costs for the Star Trek set, future iterations could expand the dataset to include older sets to track "power creep" over time, analyzing if modern cards are mathematically cheaper for the exact same abilities.

Executive Summary:

This project built a Random Forest machine learning model to deconstruct and predict the mana value formula of Magic: The Gathering cards using the Star Trek (TRK) set. Extracting live JSON data via the Scryfall API, the pipeline processed 647 mechanically unique cards, utilizing Natural Language Processing (NLP) to engineer 62 features from unstructured ability text and base stats.

The model achieved an Exact Match Rate (Accuracy) of 51.5% and a Mean Absolute Error (MAE) of 0.70. These metrics indicate that while Wizards of the Coast heavily utilizes a baseline mathematical formula to price cards (with total word count, artifact typing, and toughness being the heaviest cost determinants), nearly half of all cards receive manual Play Design adjustments to account for unquantifiable mechanics like set synergy and competitive meta balancing. The resulting interactive Tableau dashboard provides Lead Game Designers with an automated baseline pricing tool to forecast efficiency outliers and proactively prevent power creep in future set development.